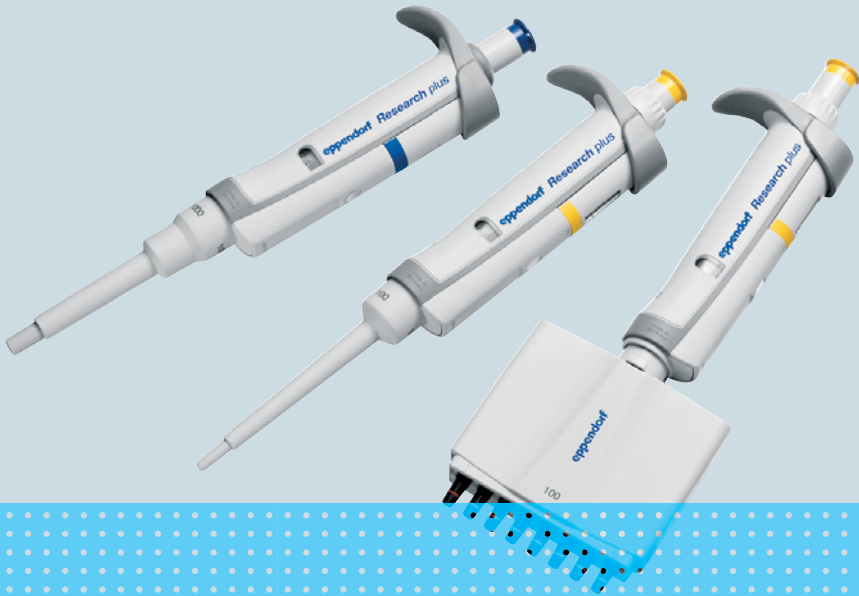


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Eppendorf Research® plus

Operating manual

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	Eppendorf Research® plus
	English (EN)

1 Operating instructions

1.1 Using this manual

- ▶ Read this operating manual completely before using the device for the first time. Observe the instructions for use of the accessories where applicable.
- ▶ This operating manual is part of the product. Please keep it in a place that is easily accessible.
- ▶ Enclose this operating manual when transferring the device to third parties.
- ▶ The current version of the operating manual for all available languages can be found on our webpage www.eppendorf.com/manuals.

1.2 Danger symbols and danger levels

1.2.1 Danger symbols

The safety instructions in this manual have the following danger symbols and danger levels:

	Biohazard		Explosive substances
	Toxic substances		Material damage
	Hazard point		

1.2.2 Danger levels

DANGER	<i>Will</i> lead to severe injuries or death.
WARNING	<i>May</i> lead to severe injuries or death.
CAUTION	May lead to light to moderate injuries.
NOTICE	May lead to material damage.

1.3 Symbols used

Depiction	Meaning
1. 2.	Actions in the specified order
▶	Actions without a specified order
•	List
<i>Text</i>	Display or software texts
	Additional information

2 **Safety**

2.1 **Intended use**

The Eppendorf Research plus pipette is designed and constructed for low-contamination transfer of liquids, in particular for samples from the human body and for reagents within the scope of an in-vitro diagnostic application in order to allow the in-vitro diagnostic medical device to be used as intended.

This pipette is an in vitro diagnostic device according to Directive 98/79/EC of the European Parliament and the Council dated October 27, 1998.

It is intended exclusively for indoor use and for operation by qualified staff.

2.2 **Warnings for intended use**



WARNING! Damage to health due to infectious liquids and pathogenic germs.

- ▶ When handling infectious liquids and pathogenic germs, observe the national regulations, the biosafety level of your laboratory, the material safety data sheets, and the manufacturer's application notes.
- ▶ Wear your personal protective equipment.
- ▶ For comprehensive regulations about handling germs or biological material of risk group II or higher, please refer to the "Laboratory Biosafety Manual" (source: World Health Organization, Laboratory Biosafety Manual, the current edition).



WARNING! Damage to health due to toxic, radioactive or aggressive chemicals.

- ▶ Wear your personal protective equipment.
- ▶ Observe the national regulations for handling these substances.
- ▶ Observe the material safety data sheets and manufacturer's application notes.



CAUTION! Danger to people due to grossly negligent use.

- ▶ Never point the opening of the device towards yourself or others.
- ▶ Only initiate liquid dispensing if it is safe to do so.
- ▶ For all dispensing tasks, make sure that you are not endangering yourself or anyone else.



CAUTION! Poor safety due to incorrect accessories and spare parts.

The use of accessories and spare parts other than those recommended by Eppendorf may impair the safety, functioning and precision of the device. Eppendorf cannot be held liable or accept any liability for damage resulting from the use of accessories and spare parts other than those recommended, or from the improper use of such equipment.

- ▶ Only use accessories and original spare parts recommended by Eppendorf.



NOTICE! Damage to device due to missing pipette tips.

- ▶ Use the pipette with fitted pipette tips only.



NOTICE! Carry-over, contamination and incorrect dispensing results due to the incorrect use of pipette tips.

The pipette tips are for single use only. Prolonged use can have a negative impact on dispensing tasks.

- ▶ Only use the pipette tips once.



NOTICE! Incorrect dispensing volume with special liquids and due to temperature differences.

Solutions with physical data very different to that of water or temperature differences between the pipette, pipette tip and liquid may result in incorrect dispensing volumes.

- ▶ Avoid temperature differences between the pipette, pipette tip and liquid.



NOTICE! Damage to device due to penetration of liquid.

- ▶ Only immerse the pipette tip in the liquid.
- ▶ Do not put the pipette down when the pipette tip is filled.
- ▶ The pipette itself may not come into contact with the liquid.

3 Product description

3.1 Delivery package

Quantity	Description
1	Research plus
1	Adjustment tool (Allen key with a blue handle)
5	Red adjustment seal
1	Pin (remove the safety plug)
1	Operating manual
1	Certificate

3.1.1 Single-channel pipettes

Quantity	Description
1	Locking ring ($\leq 1000\ \mu\text{L}$)
10	Protection filter (2.5 mL – 10 mL)
1	Pipette wrench (2.5 mL – 10 mL)

3.1.2 Multi-channel pipettes

Quantity	Description
1	Multi-channel tool 100/300 (100 μL and 300 μL)
1	Multi-channel tool 1200 (1200 μL)
1	Unlocking tool (1200 μL)
2	Locking clip (8-channel lower part with 10 μL , 100 μL and 300 μL)
3	Locking clip (12-channel lower part with 10 μL , 100 μL and 300 μL)

3.2 Features

The Eppendorf Research plus pipette is a piston-stroke pipette for aspirating and dispensing liquids. The pipette operates according to the air cushion principle. A suitable pipette tip must be fitted prior to use. The dispensing and volume setting ring are controlled with the dispensing button. The pipette tips are ejected with the separate ejector. Depending on the model, volumes between 0.1 µL and 10 mL can be dispensed.

3.2.1 Pipette models

Different variants are available:

- Single-channel pipettes with fixed volume setting
- Single-channel pipettes with variable volume setting
- Multi-channel pipettes with 8 or 12 channels with fixed cone spacing and variable volume setting
- Multi-channel pipettes with 16 or 24 channels with fixed cone spacing (4.5 mm) and variable volume setting

3.3 Product overview

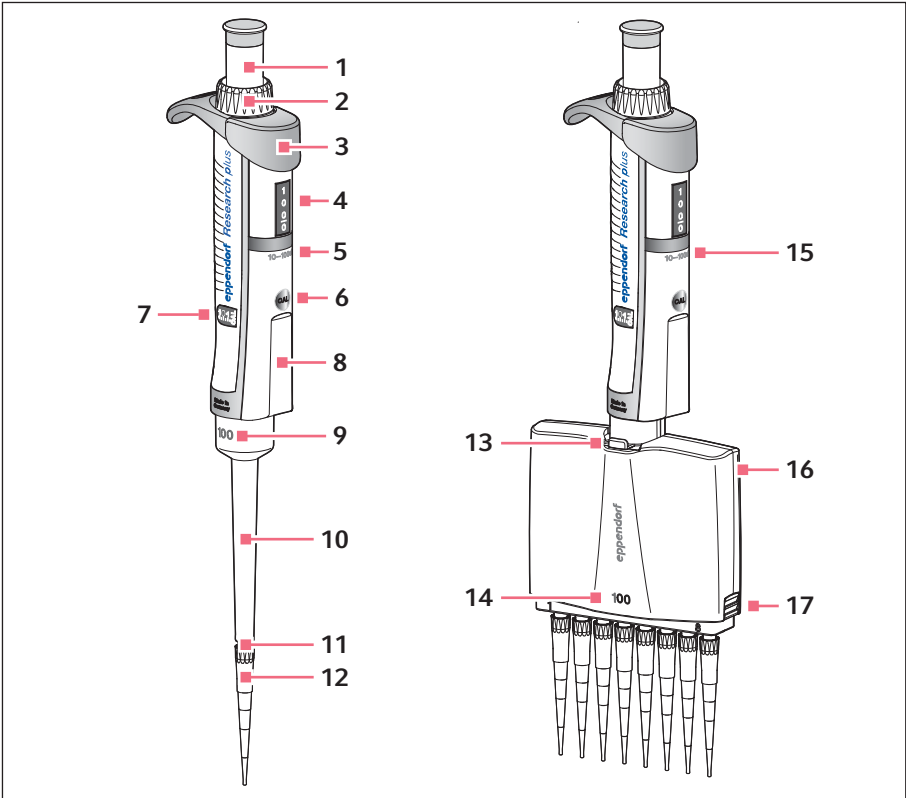


Fig. 3-1: Single-channel pipette and multi-channel pipette

- | | |
|---|---|
| 1 Dispensing button | 10 Ejector sleeve |
| 2 Volume setting ring | 11 Tip cone |
| 3 Ejector | 12 Pipette tip |
| 4 Volume display | 13 Lever |
| 5 Single-channel upper part with nominal volume | 14 Multi-channel lower part with nominal volume |
| 6 Adjustment opening | 15 Multi-channel upper part with nominal volume |
| 7 Adjustment display | 16 Housing lid |
| 8 Labeling area | 17 Latch |
| 9 Single-channel lower part with nominal volume | Opening the lower part |

3.3.1 Multi-channel lower part with fixed cone spacing

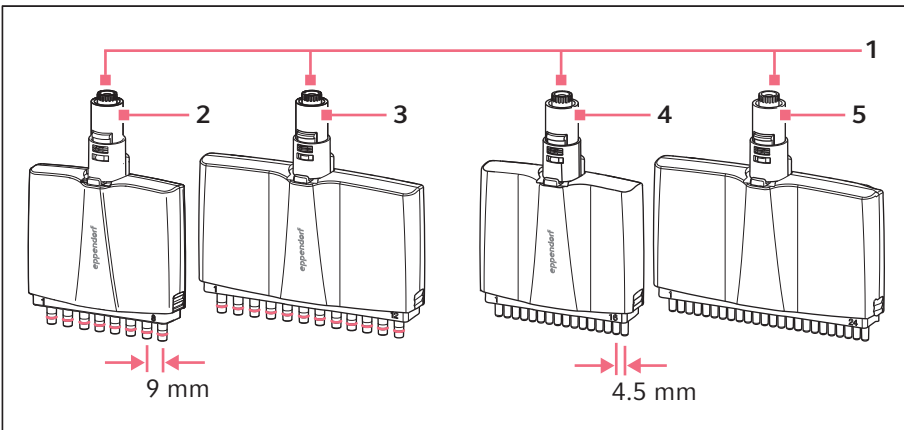


Fig. 3-2: Multi-channel lower parts with fixed cone spacing

- | | |
|--------------------------------------|---|
| 1 Mechanical coupling | 4 16-channel lower part |
| 2 8-channel lower part | 4.5 mm cone spacing for 384-well plates |
| 9 mm cone spacing for 96-well plates | 5 24-channel lower part |
| 3 12-channel lower part | 4.5 mm cone spacing for 384-well plates |
| 9 mm cone spacing for 96-well plates | |

3.4 Materials



NOTICE! Aggressive substances may damage components, consumables and accessories.

- ▶ Check the chemical resistance before using organic solvents or aggressive chemicals.
- ▶ Only use liquids whose vapors do not attack the materials used.

The components of the pipette which can be accessed by the user are made of the following materials:

Assembly	Material
External surfaces of the upper part	• Improved polypropylene (PP) • Polycarbonate (PC) • Polyetherimide (PEI) • Foil
Viewing window	• Polycarbonate (PC)
Exterior and interior of lower parts	• Improved polypropylene (PP) • Polyvinylidene fluoride (PVDF) • Polyetherimide (PEI) • Polyphenylene sulfide (PPS) • Polyetheretherketone (PEEK) • Polytetrafluoroethylene (PTFE) • Ethylene propylene diene rubber (EPDM) • Silicone • Steel (stainless steel and spring steel)



You can find information on the resistance to chemicals on our webpage www.eppendorf.com/manuals.

3.5 Warranty

In case of warranty claims, contact your local Eppendorf contractual partner.

No warranty is given in the following cases:

- In the case of misuse.
- If unauthorized persons open the upper part.

The following assemblies are excluded from the warranty:

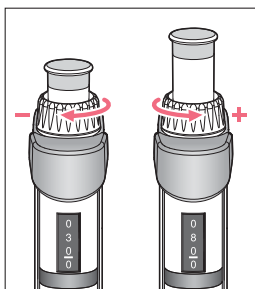
- Wear parts

4 Operation

4.1 Setting the volume

4.1.1 Setting a small volume

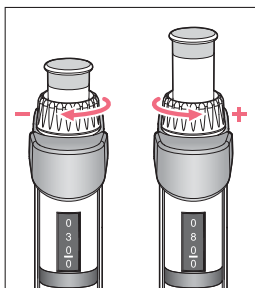
- i** Set the volume from a high to a low value.



1. Turn the volume setting ring clockwise. The set volume is shown on the volume display.

4.1.2 Setting a large volume

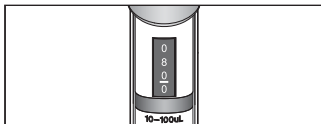
- i** Set the volume from a high to a low value.



1. Turn the volume setting ring counterclockwise. The set volume is shown on the volume display.

4.1.3 Reading the set volume

The volume is shown on the volume display. The decimal place is under the hyphen.

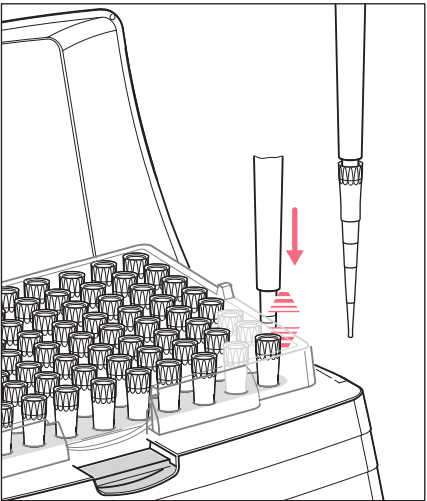


1. Read the volume from top to bottom. Set volume: 80 µL.

4.2 Attaching the pipette tips

The pipette tip can be attached by hand or mounted directly from a tip holder (tray) with the pipette.

-  The pipette tips are single-use items.
-  The control button and the trays are color-coded. The color marks the pipette size and the volume of the pipette tip (epT.I.P.S.).



1. Insert the tip cone into the pipette tip using light force.

4.3 Optimum immersion depths

Volume	Immersion depth
0.1 µL – 1 µL	1 mm
1 µL – 100 µL	2 – 3 mm
100 µL – 1000 µL	2 – 4 mm
1 mL – 10 mL	3 – 5 mm

4.4 Forward pipetting

4.4.1 Aspirating liquid

Prerequisites

- The pipette tip has been attached.



To ensure maximum precision and accuracy, we recommend initially wetting each new tip by aspirating and dispensing the liquid one to three times.

1. Press down the dispensing button to the first stop.
2. Immerse the pipette tip vertically in the liquid.
3. Maintain the immersion depth and let the dispensing button slide back slowly. Liquid is aspirated into the pipette tip.
4. Wait until the liquid has been aspirated.
5. Remove the pipette tip from the liquid.



If necessary, wipe the pipette tip against the tube inner wall.

4.4.2 Dispensing liquid

1. Place the pipette tip on the tube inner wall at a steep angle.
2. Slowly press down the dispensing button to the first stop. The liquid is dispensed.
3. Wait until the flow of liquid stops.
4. Press down the dispensing button to the second stop. The pipette tip is completely emptied.
5. Hold down the dispensing button and wipe the pipette tip on the tube inner wall.

4.4.3 Ejecting the pipette tip

- ▶ Press the ejector.
The pipette tip is ejected.

4.5 Reverse pipetting

An additional volume (blow out) is aspirated when performing reverse pipetting. This can improve dispensing results for viscous or foaming liquids. Volume limits may occur when using filter tips.

4.5.1 Aspirating liquid

1. Press down the dispensing button to the second stop.
2. Immerse the pipette tip vertically in the liquid.
3. Maintain the immersion depth and let the dispensing button slide back slowly.
Liquid is aspirated into the pipette tip.
4. Wait until the liquid has been aspirated.
5. Remove the pipette tip from the liquid.



If necessary, wipe the pipette tip against the tube inner wall.

4.5.2 Dispensing liquid

1. Place the pipette tip on the tube inner wall at a steep angle.
2. Slowly press down the dispensing button to the first stop.
The liquid is dispensed.
3. Wait until the flow of liquid stops.
4. Hold down the dispensing button and wipe the pipette tip on the tube inner wall.
Residual liquid remains in the pipette tip.
During dispensing, the additional volume (blow out) does not belong to the dispensing volume.

4.5.3 Ejecting the pipette tip

1. Press down the dispensing button to the second stop.
The residual liquid is dispensed.
The residual liquid can be discarded.
2. Press the ejector.
The pipette tip is ejected.

4.6 Storing the pipette

The pipette can be stored in a pipette carousel, a wall mount or in horizontal position.

5 Troubleshooting

5.1 Error search

Problem	Cause	Solution
The adjustment seal has been removed; the adjustment display has been changed.	The pipette has been adjusted for a different liquid.	► Adjust the pipette for the liquid used.

5.1.1 Control button

Problem	Cause	Solution
The control button jams and does not move smoothly.	• The piston is contaminated. • The seal is contaminated. • The seal is defective. • The pipette is blocked.	► Clean the lower part. ► Replace the protection filter (2.5 mL – 10 mL).

5.1.2 Dispensing

Problem	Cause	Solution
Liquid is dripping from the tip and/or the dispensed volume is incorrect.	Pipette tip is loose.	► Firmly attach the pipette tip ► Deactivate the spring-loaded action. ► Use epT.I.P.S. pipette tips. ► For 2.5 mL, 5 mL and 10 mL ep Dualfilter T.I.P.S. pipette tips, work without a protection filter in the pipette.
	Liquid with high vapor pressure and/or different density.	► Wet the tip several times and adjust the pipette for the liquid used.
	Pipetting was too quick.	► Move the control button slowly.
	The tip is withdrawn from the liquid too quickly.	► Withdraw the tip slowly and with a time delay (approx. 3 seconds) from the liquid.
	Liquid aspirated with blow out and dispensed with blow out.	► Repeat dispensing correctly.

Problem	Cause	Solution
	The piston is contaminated.	► Clean and grease the piston.
	The tip cone is damaged.	► Replace the lower part or channel.
	The O-rings of the tip cones are damaged.	► Replace the O-rings (100 µL, 300 µL and 1200 µL multi-channel).

5.1.3 Pipette tip

Problem	Cause	Solution
Pipette tip is loose.	The pipette tip is not compatible.	► Use epT.I.P.S. pipette tips. ► Use a suitable size.
	Higher attachment forces are required.	► Firmly attach the pipette tip. ► Deactivate the spring-loaded action.
Liquid is dripping from the tip and/or the dispensed volume is incorrect.	The piston is damaged.	► Replace the piston.

5.1.4 Tip cone

Problem	Cause	Solution
No spring-loaded action of the tip cone.	The spring-loaded action is blocked.	► Remove the locking ring (single-channel). ► Remove the locking clip (multi-channel).
	Use of a 2.5 mL-, 5 mL-, 10 mL or 1200 µL pipette.	The tip cone of these sizes does not have spring-loading action.

To prevent dispensing errors, regularly check the precision and accuracy of the pipette.

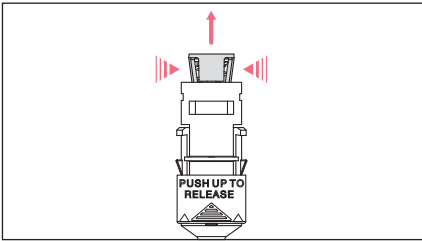


The gravimetric test and the conversion of the measured values for the volume are described in the document *“Standard operating procedure for manual dispensing systems”*. The document is available on the webpage www.eppendorf.com/manuals.

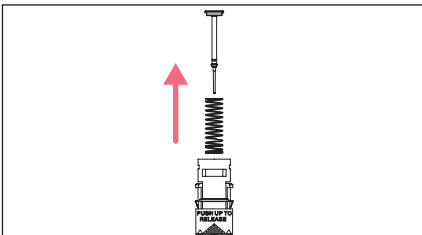
6.1.2 Disassembling the lower part

Prerequisites

- The ejector sleeve has been removed.
- The lower part has been removed from the upper part.



1. Slightly squeeze the stop pins at the piston mount.
2. Remove the piston mount.



3. Remove the piston spring and the piston.
4. Remove the piston from the piston spring (not possible for pipettes with blue dispensing button).

6.2 Disassembling the single-channel pipette ≥ 2.5 mL

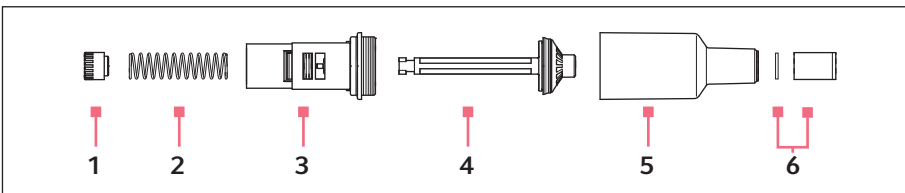
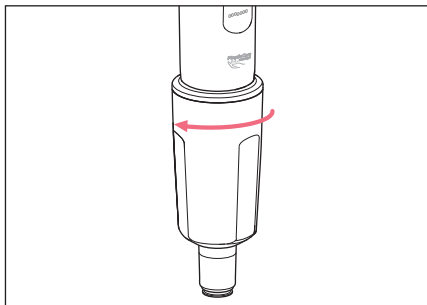


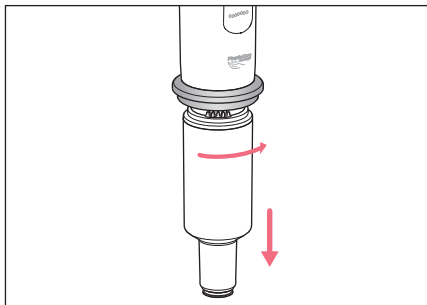
Fig. 6-2: Single-channel lower part ≥ 2.5 mL

- | | |
|-----------------|--|
| 1 Piston mount | 4 Piston |
| 2 Piston spring | 5 Cylinder with tip cone |
| 3 Piston guide | 6 Filter sleeve with protection filter |

6.2.1 Removing the lower part



1. Unscrew the ejector sleeve.

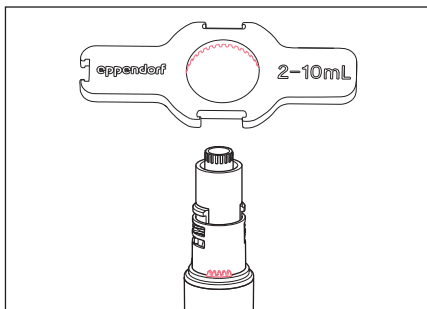


2. Keep the dispensing button fully pressed.
3. Turn the lower part approximately 30° to the right.
The lower part is released from the upper part.

6.2.2 Disassembling the lower part

Prerequisites

- The lower part is removed.



1. Fit the pipette wrench onto the lower part.
2. Hold the cylinder and unscrew it from the lower part.

6.3 Assembling the single-channel pipette $\leq 1000 \mu\text{L}$

1. Insert the piston spring.



If using piston springs with double coil, this must point down.

2. Carefully insert the piston into the cylinder from the top.
3. Compress the piston spring with piston and hold.
4. Squeeze the stop pins at the piston mount and attach them.
5. Press on the piston from above and check for free movement.
The piston must be able to move freely without resistance.
6. Insert the lower part into the upper part until it engages.
7. Fit the ejector sleeve.

6.4 Assembling the single-channel pipette $\geq 2 \text{ mL}$

6.4.1 Assembling the lower part

1. Insert the piston into the piston guide.
2. Insert the piston spring into the piston guide.
3. Attach the piston mount and press the piston spring into the piston guide.
4. Turn the piston mount by 90° and let it engage.
5. Insert the lower part into the upper part until it engages.
6. Attach and tighten the ejector sleeve.

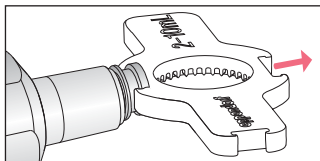
6.4.2 Checking the function

This inspection ensures that the pipette has been assembled correctly.

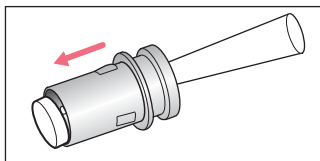
- Carry out a gravimetric test of the systematic and random errors.

6.5 Replacing the protection filter ≥ 2 mL

The protection filter in the tip cone must be replaced after every contact with liquids.

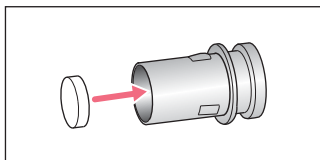


1. Place a pipette wrench with the appropriate opening onto the filter sleeve.
2. Pull out the filter sleeve.

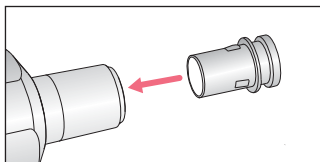


3. Use a pipette tip to push out the filter.

4. Clean the filter sleeve.



5. Insert a new protection filter into the filter sleeve.

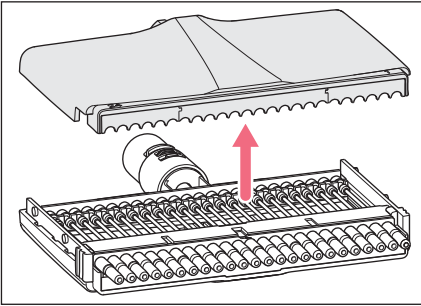


6. Insert the filter sleeve in the tip cone.

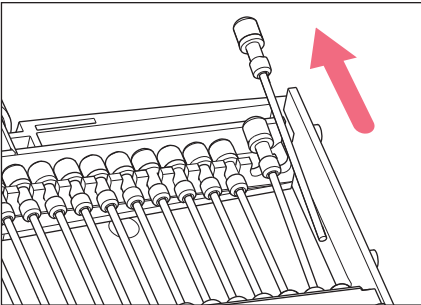
6.6 Disassembling the multi-channel lower part – 4.5 mm cone spacing**6.6.1 Opening the multi-channel lower part**

Prerequisites

- The lower part is detached from the pipette upper part.



1. Slide both latches on the sides of the lower part down.
2. Remove the housing lid.

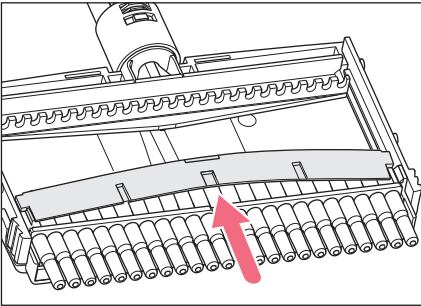
6.6.2 Removing pistons

1. Lift the piston out of the piston mount.
2. Pull the piston out of the cylinder.

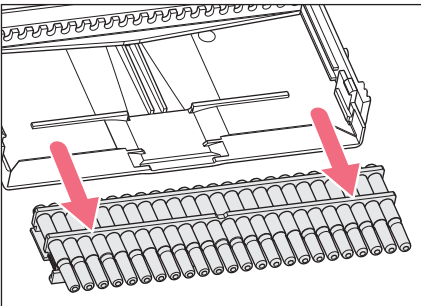
6.6.3 Removing the cylinder block

Prerequisites

- All pistons are removed.



1. Lift the clamping bar at its center and remove it.



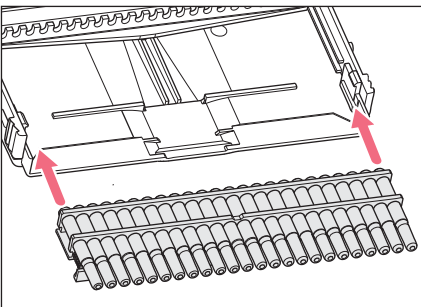
2. Slide the cylinder block straight down and out of the housing case.
The lower part can be cleaned.

6.7 Assembling the multi-channel lower part – 4.5 mm cone spacing

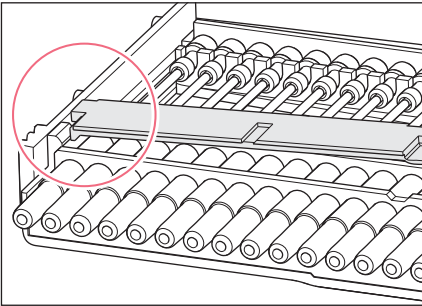
6.7.1 Inserting the cylinder block

Prerequisites

- All pistons are removed.

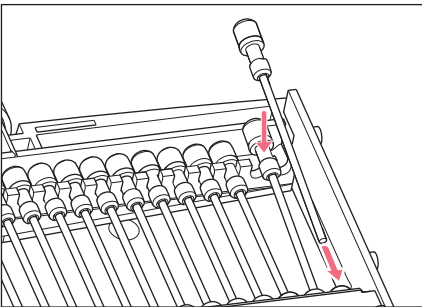


1. Insert the cylinder block straight into the housing case.
2. Slide the cylinder block straight into the housing case.
The cylinder block needs to be flush with the edge of the housing case.



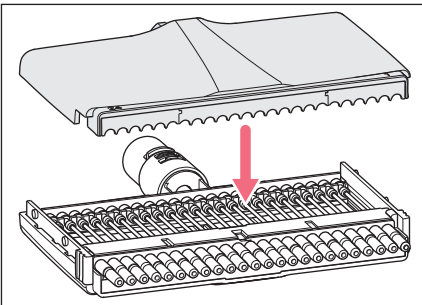
3. Insert the clamping bar on one side into the housing case under the lug.
4. Bend the clamping bar and slide it under the lug on the other side.
The cylinder block is now fixed.

6.7.2 Insert the pistons



1. Slide the piston into the cylinder.
2. Insert the end of the piston into the piston mount.

6.7.3 Closing the multi-channel lower part



1. Attach the housing lid.
2. Slide both latches on the sides of the lower part up.

6.8 Replacing O-rings – multi-channel lower part

Replace worn or damaged O-rings on multi-channel lower parts.

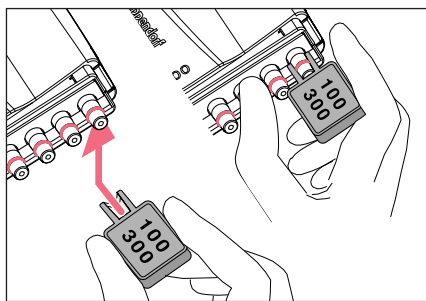
Applies to multi-channel lower parts:

- 100 µL
- 300 µL
- 1200 µL

6.8.1 Removing the O-ring

Prerequisites

- O-ring tool (included in the delivery package)



1. Push the opening of the O-ring tool against the tip cone.
2. Supporting the O-ring tool with your thumb, push it against the tip cone. The O-ring is cut at one location.
3. Remove the O-ring tool and O-ring.

6.8.2 Mounting a new O-ring – 100 µL and 300 µL

Prerequisites

- Mounting aid (shortened pipette tip)

1. Fit the installation aid on the tip cone.
2. Push the O-ring over the installation aid and onto the tip cone. The O-ring must be positioned in the groove in the tip cone.
3. Remove the installation aid
4. Attach the pipette tip and check it is seated correctly. The pipette tip must be firmly seated on the tip cone.

6.8.3 Mounting a new O-ring – 1200 µL

1. Push the O-ring onto the tip cone. The O-ring must be positioned in the groove in the tip cone.
2. Attach the pipette tip and check it is seated correctly. The pipette tip must be firmly seated on the tip cone.

6.9 Adjusting the pipette



The execution of changes to the user adjustment and factory adjustment is described on our website www.eppendorf.com/manuals.

6.10 Cleaning



NOTICE! Damage to device from unsuitable cleaning agents or sharp or pointed objects.

Unsuitable cleaning agents may damage the device.

- ▶ Do not use corrosive cleaning agents, strong solvents, or abrasive polishes.
- ▶ Check the compatibility with the materials used.
- ▶ Please note the information on chemical resistance.
- ▶ Do **not** clean the device with acetone or organic solvents with a similar effect.
- ▶ Do **not** use sharp or pointed objects to clean the device.



NOTICE! Damage to device due to penetration of liquid.

- ▶ Only immerse the pipette tip in the liquid.
 - ▶ Do not put the pipette down when the pipette tip is filled.
 - ▶ The pipette itself may not come into contact with the liquid.
-

6.10.1 Cleaning and disinfecting the pipette

All single and multi-channel lower parts are wear parts. Clean them after contamination, use of aggressive chemicals and/or heavy stress. If the lower parts are worn or damaged, replace these parts.

1. Moisten a cloth with a cleaning agent and a decontamination agent.
2. Remove external contamination.
3. Moisten a new cloth with water.
4. Wipe down the housing.

6.10.2 Cleaning and disinfecting the lower part

Prerequisites

- Heavy contamination caused by ingress of liquid must be removed.
 - Lower part has been removed and disassembled.
1. Remove piston grease.
 2. Rinse the lower part with cleaning agent or decontamination agent or let it soak.
 -  Observe the contact time recommended by the manufacturer.
 3. Thoroughly rinse the lower part with demineralized water.
 4. Let it dry.
 5. Lubricate the piston or cylinder.
 -  See instructions for use "Grease for pipettes".
 6. Assemble the lower part.

6.10.3 Sterilizing the pipette with UV light

The pipette can be sterilized with UV light at 254 nm.

6.11 Autoclaving the pipette



NOTICE! Damage to device due to incorrect handling.

- ▶ Do not use any additional disinfectants, decontamination agents or sodium hypochlorite during autoclaving or UV irradiation.

6.11.1 Autoclaving



Autoclave the filter sleeve and the protection filter separately.



The upper part and lower part can be autoclaved in assembled state. The lower part does not have to be disassembled.

Prerequisites

- The pipette has been cleaned.
 - Any residual cleaning agent has been removed.
 - The protection filter has been removed.
1. Autoclave the pipette at 121°C and 1 bar overpressure for 20 minutes.
 2. Cool the pipette to ambient temperature and let it dry.
 -  For maximum precision and accuracy, a gravimetric test is recommended after autoclaving.
-  The piston does **not** need to be greased following autoclaving.

6.12 Decontamination before shipment



CAUTION! Use of a contaminated device may result in personal injury and damage to the device.

- ▶ Clean and decontaminate the device in accordance with the cleaning instructions before shipping or storage.
-

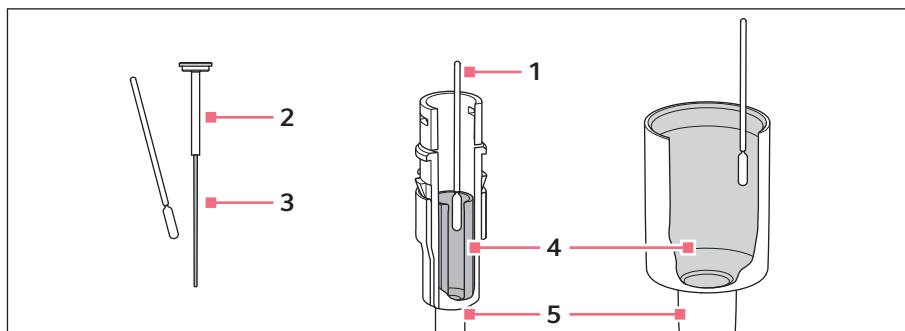
Hazardous substances are:

- solutions presenting a hazard to health
 - potentially infectious agents
 - organic solvents and reagents
 - radioactive substances
 - proteins presenting a hazard to health
 - DNA
1. Please note the information in the document "Decontamination certificate for product returns".
It is available as PDF document on our website www.eppendorf.com/decontamination.
 2. Enter the serial number of the device in the decontamination certificate.
 3. Enclose the completed decontamination certificate for returned goods with the device.
 4. Send the device to Eppendorf AG or an authorized service center.

6.13 Lubricating the piston or cylinder

The piston or the cylinder in the lower part of the pipette must be relubricated after cleaning or after decontamination.

i The special grease can be ordered as an accessory. The ordering information can be found on our webpage www.eppendorf.com/manuals



1 Cotton swab

2 Piston
≤ 20 µL

3 Running surface

4 Cylinder

5 Lower part
> 20 µL

6.13.1 Lubricating the piston

Prerequisites

- For volumes ≤ 20 µL.
- Lower part is disassembled.

1. Place a small amount of grease on the cotton swab.
2. Apply a thin film of grease to the running surface of the piston.
The lower part can now be reassembled.

6.13.2 Lubricating the cylinder

Prerequisites

- For volumes > 20 µL.
- Lower part is disassembled.

1. Place a small amount of grease on the cotton swab.
2. Apply a thin film of grease to the inner wall of the piston.
The lower part can now be reassembled.

7 Technical data

7.1 Adjustable sub-steps – single-channel pipettes

Model	Increment
0.1 µL – 2.5 µL	0.002 µL
0.5 µL – 10 µL	0.01 µL
2 µL – 20 µL	0.02 µL
10 µL – 100 µL	0.1 µL
20 µL – 200 µL	0.2 µL
30 µL – 300 µL	0.2 µL
100 µL – 1000 µL	1 µL
0.25 mL – 2.5 mL	2 µL
0.5 mL – 5 mL	0.005 mL
1 mL – 10 mL	0.01 mL

7.2 Adjustable sub-steps – multi-channel pipettes

Model	Increment
0.5 µL – 10 µL	0.01 µL
1 µL – 20 µL	0.02 µL
5 µL – 100 µL	0.1 µL
10 µL – 100 µL	0.1 µL
30 µL – 300 µL	0.2 µL
120 µL – 1200 µL	1 µL

7.3 Ambient conditions

	Temperature range	Relative humidity
Storage without transport package	-5 °C – 45 °C	10 % – 95 %
Operating conditions	5 °C – 40 °C	10 % – 95 %

8 Measurement deviations according to Eppendorf AG**8.1 Single-channel pipettes with fixed volume setting**

Model	Test tip epT.I.P.S.	Error of measurement			
		systematic		random	
		± %	± µL	± %	± µL
10 µL medium gray	0.1 µL – 20 µL medium gray 40 mm	1.2	0.12	0.6	0.06
20 µL light gray	0.5 µL – 20 µL L light gray 46 mm	0.8	0.16	0.3	0.06
10 µL yellow	2 µL – 200 µL yellow 53 mm	1.2	0.12	0.6	0.06
20 µL yellow		1.0	0.2	0.3	0.06
25 µL yellow		1.0	0.25	0.3	0.08
50 µL yellow		0.7	0.35	0.3	0.15
100 µL yellow		0.6	0.6	0.2	0.2
200 µL yellow		0.6	1.2	0.2	0.4
200 µL blue	50 µL – 1000 µL blue 71 mm	0.6	1.2	0.2	0.4
250 µL blue		0.6	1.5	0.2	0.5
500 µL blue		0.6	3.0	0.2	1.0
1000 µL blue		0.6	6.0	0.2	2.0

8.2 Single-channel pipettes with variable volume setting

Model	Test tip epT.I.P.S.	Testing volume	Error of measurement			
			systematic		random	
			± %	± µL	± %	± µL
0.1 µL – 2.5 µL dark gray	0.1 µL – 10 µL dark gray 34 mm	0.1 µL	48	0.048	12	0.012
		0.25 µL	12	0.03	6.0	0.015
		1.25 µL	2.5	0.031	1.5	0.019
		2.5 µL	1.4	0.035	0.7	0.018
0.5 µL – 10 µL medium gray	0.1 µL – 20 µL medium gray 40 mm	0.5 µL	8.0	0.04	5.0	0.025
		1 µL	2.5	0.025	1.8	0.018
		5 µL	1.5	0.075	0.8	0.04
		10 µL	1.0	0.1	0.4	0.04
2 µL – 20 µL light gray	0.5 µL – 20 µL L light gray 46 mm	2 µL	5.0	0.1	1.5	0.03
		10 µL	1.2	0.12	0.6	0.06
		20 µL	1.0	0.2	0.3	0.06
2 µL – 20 µL yellow	2 µL – 200 µL yellow 53 mm	2 µL	5.0	0.1	1.5	0.03
		10 µL	1.2	0.12	0.6	0.06
		20 µL	1.0	0.2	0.3	0.06
10 µL – 100 µL yellow	2 µL – 200 µL yellow 53 mm	10 µL	3.0	0.3	1.0	0.1
		50 µL	1.0	0.5	0.3	0.15
		100 µL	0.8	0.8	0.2	0.2
20 µL – 200 µL yellow	2 µL – 200 µL yellow 53 mm	20 µL	2.5	0.5	0.7	0.14
		100 µL	1.0	1.0	0.3	0.3
		200 µL	0.6	1.2	0.2	0.4
30 µL – 300 µL orange	20 µL – 300 µL orange 55 mm	30 µL	2.5	0.75	0.7	0.21
		150 µL	1.0	1.5	0.3	0.45
		300 µL	0.6	1.8	0.2	0.6
100 µL – 1000 µL blue	50 µL – 1000 µL blue 71 mm	100 µL	3.0	3.0	0.6	0.6
		500 µL	1.0	5.0	0.2	1.0
		1000 µL	0.6	6.0	0.2	2.0
0.25 mL – 2.5 mL red	0.25 mL – 2.5 mL red 115 mm	0.25 mL	4.8	12	1.2	3
		1.25 mL	0.8	10	0.2	2.5
		2.5 mL	0.6	15	0.2	5
0.5 mL – 5 mL violet	0.1 mL – 5 mL violet 120 mm	0.5 mL	2.4	12	0.6	3
		2.5 mL	1.2	30	0.25	6.25
		5.0 mL	0.6	30	0.15	7.5

Model	Test tip epT.I.P.S.	Testing volume	Error of measurement			
			systematic		random	
			± %	± µL	± %	± µL
1 mL – 10 mL turquoise	1 mL – 10 mL turquoise 165 mm	1.0 mL	3.0	30	0.6	6
		5.0 mL	0.8	40	0.2	10
		10.0 mL	0.6	60	0.15	15

8.3 Multi-channel pipettes with fixed cone spacing

Model	Test tip epT.I.P.S.	Testing volume	Error of measurement			
			systematic		random	
			± %	± µL	± %	± µL
0.5 µL – 10 µL medium gray 8/12-channel	0.1 µL – 20 µL medium gray 40 mm	0.5 µL	12	0.06	8.0	0.04
		1 µL	8.0	0.08	5.0	0.05
		5 µL	4.0	0.2	2.0	0.1
		10 µL	2.0	0.2	1.0	0.1
1 µL – 20 µL pearl white 16/24-channel	1 µL – 20 µL pearl white 42 mm	1 µL	12	0.12	8	0.08
		2 µL	8	0.16	5	0.1
		10 µL	4	0.4	2	0.2
		20 µL	2	0.4	1	0.2
5 µL – 100 µL light yellow 16/24-channel	5 µL – 100 µL light yellow 53 mm	5 µL	6	0.3	4	0.2
		10 µL	3	0.3	2	0.2
		50 µL	1.2	0.6	0.8	0.4
		100 µL	1	1	0.6	0.6
10 µL – 100 µL yellow 8/12-channel	2 µL – 200 µL yellow 53 mm	10 µL	3.0	0.3	2.0	0.2
		50 µL	1.0	0.5	0.8	0.4
		100 µL	0.8	0.8	0.3	0.3
30 µL – 300 µL orange 8/12-channel	20 µL – 300 µL orange 55 mm	30 µL	3.0	0.9	1.0	0.3
		150 µL	1.0	1.5	0.5	0.75
		300 µL	0.6	1.8	0.3	0.9
120 µL – 1200 µL dark green 8/12-channel	50 µL – 1250 µL dark green 103 mm	120 µL	6.0	7.2	0.9	1.08
		600 µL	2.7	16.2	0.4	2.4
		1200 µL	1.2	14.4	0.3	3.6

8.4 Test conditions

Test conditions and test evaluation in compliance with ISO 8655, Part 6. Tested using a standardized analytical balance with evaporation protection.



The three largest testing volumes per tip (10 %, 50 %, 100 % of the nominal volume) correspond to the specifications in accordance with ISO 8655, Part 2 or Part 5. The test is to be carried out with these three testing volumes for testing of the systematic and random error in compliance with the standard. The smallest adjustable volume serves to provide additional information.

- Number of determinations per volume: 10
- Water according to ISO 3696
- Inspection at 20°C – 27°C
Maximum temperature variation during measurement $\pm 0.5^\circ\text{C}$
- Dispensing onto the tube inner wall
- Forward pipetting

8.5 Special tips for single-channel pipettes

Model	Special tip epT.I.P.S. Long	Testing volume	Error of measurement			
			systematic		random	
			$\pm \%$	$\pm \mu\text{L}$	$\pm \%$	$\pm \mu\text{L}$
0.5 mL – 5 mL violet	0.1 mL – 5 mL L violet 175 mm	0.5 mL	5.0	25	1.0	5
		2.5 mL	3.0	75	0.9	22.5
		5.0 mL	2.0	100	0.8	40
1 mL – 10 mL turquoise	1 mL – 10 mL L turquoise 243 mm	1.0 mL	6.0	60	1.0	10
		5.0 mL	3.0	150	0.9	45
		10.0 mL	2.0	200	0.7	70

9 Ordering information

9.1 Single-channel pipettes with fixed volume setting

Order no. (International)	Description
	Eppendorf Research plus single-channel, fixed
3121 000.015	10 µL, medium gray
3121 000.023	10 µL, yellow
3121 000.031	20 µL, light gray
3121 000.040	20 µL, yellow
3121 000.058	25 µL, yellow
3121 000.066	50 µL, yellow
3121 000.074	100 µL, yellow
3121 000.082	200 µL, yellow
3121 000.090	200 µL, blue
3121 000.104	250 µL, blue
3121 000.112	500 µL, blue
3121 000.120	1000 µL, blue

9.2 Single-channel pipettes with variable volume setting

Order no. (International)	Description
	Eppendorf Research plus Single-channel, variable
3120 000.011	0.1 – 2.5 µL, dark gray
3120 000.020	0.5 – 10 µL, medium gray
3120 000.097	2 – 20 µL, light gray
3120 000.038	2 – 20 µL, yellow
3120 000.046	10 – 100 µL, yellow
3120 000.054	20 – 200 µL, yellow
3120 000.100	30 – 300 µL, orange
3120 000.062	100 – 1000 µL, blue
3120 000.143	0.25 – 2.5 mL, red
3120 000.070	0.5 – 5 mL, violet
3120 000.089	1 – 10 mL, turquoise

Ordering information

Eppendorf Research® plus
English (EN)

9.3 Multi-channel pipettes with fixed cone spacing

9.3.1 9 mm cone spacing for 96-well plates

Order no. (International)	Description
	Eppendorf Research plus 8-channel
3122 000.019	0.5 – 10 µL, medium gray
3122 000.035	10 – 100 µL, yellow
3122 000.051	30 – 300 µL, orange
3122 000.213	120 – 1200 µL, dark green
	Eppendorf Research plus 12-channel
3122 000.027	0.5 – 10 µL, medium gray
3122 000.043	10 – 100 µL, yellow
3122 000.060	30 – 300 µL, orange
3122 000.221	120 – 1200 µL, dark green

9.3.2 4.5 mm cone spacing for 384-well plates

Order no. (International)	Description
	Eppendorf Research plus 16-channel
3122 000.078	1 – 20 µL, cone spacing 4,5 mm
3122 000.094	5 – 100 µL, cone spacing 4,5 mm
	Eppendorf Research plus 24-channel
3122 000.086	1 – 20 µL, cone spacing 4,5 mm
3122 000.108	5 – 100 µL, cone spacing 4,5 mm



The ordering information can be found on our website
www.eppendorf.com/manuals.

Declaration of Conformity

The product named below fulfills the requirements of directives and standards listed. In the case of unauthorized modifications to the product or an unintended use this declaration becomes invalid.

Product name:

Eppendorf Research® plus 3120, 3121, 3122

Product type:

fixed- volume, variable- volume, multichannel pipette

Relevant directives / standards:

98/79/EC EN 14971, EN 18113- 3, EN 980, EN 15223- 1, EN 62366

EN ISO 8655- 1, EN ISO 8655- 2, EN ISO 8655- 6

Date: February 16, 2016



Management Board



Portfolio Management

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